Assessments

MATH 107: Introduction to Linear Algebra — Fall 2026

Labs

Lab points (effort) are earned in class: work the hub exercises in Octave or Python, handwrite answers on the one-pager hard copy, and get instructor initials after a short check-in. Exercises are on each lab page in both languages; the one-pager is linked from the hub so you can preview it. Quizzes will test lab ideas. See Lab 0 for the pattern.

Lab	Date	Learning objectives
Lab 0: Computers Speak Vectors	Fri Aug 28	Form and interpret linear combinations; enter, index, and modify vectors in Octave or Python; plot and debug supplied code.
Lab 1: Finding Similar Things	Fri Sep 18	Short (~15 min after Quiz 3): compute distances; name a nearest neighbor; see the effect of rescaling one feature.
Lab 2: Finding Similar Things (software)	Mon Sep 28	Software practice on Lectures 08–10 (distances, nearest neighbor, scale, optional k-means assignment). Quiz 4 was Fri Sep 25.
Lab 3: Transforming Space	Fri Oct 9	Predict and apply 2D/3D matrix transformations; plot before/after; adapt a short program.
Lab 4: Predict a System Through Time	Fri Oct 23	Implement $x_{t+1} = Ax_t$; iterate; hand-check first steps; compare parameters; record unexplained long-run behavior.
Micro-lab: computational cost	Fri Oct 30	Compare equivalent computation orders; count storage; measure runtime; justify an order quantitatively.
Lab 5: Why Did the System Behave That Way?	Fri Nov 20	Compute eigenpairs with software; verify $Av \approx \lambda v$; explain a simulation; interpret complex eigenvalues.
Lab 6: Impossible Equations	Fri Dec 4	Compute a least-squares fit; plot residuals and RMS; check orthogonality to columns of A; interpret usefulness.
Lab 7: Capstone	Fri Dec 4 – Wed Dec 9	Low-rank strategy, error/storage/runtime, and communication

Labs (and quizzes) are scheduled on **Fridays**.

Quizzes

Ten quizzes, in the first 10 minutes of class. The lowest two scores are dropped. Practice packets are grouped as **Core** and **Stretch**. Each Friday quiz covers **only the prior instructional week** (not that day's new lecture or lab).

A consistent quiz has three parts:

- **Fluency** (4–5 points): one small calculation closely modeled on a practice item.
- **Meaning** (3–4 points): explain or interpret what the calculation says.
- **Computation/application** (2–3 points): interpret code, a plot, a residual, or software output.

You will not write substantial code from scratch on a timed quiz. Typical prompts: complete one missing line, predict three lines of output, identify an error, interpret output, or choose an appropriate command.

Every second or third quiz includes a short cumulative item from an earlier packet (beginning with Quiz 3).

Source key

Code	Source
B 3.10	Boyd and Vandenberghe, Exercise 3.10
S 6.1 #10	Strang, Section 6.1, Problem 10
L Ch. 2 Ex. 3	Lachniet, *Introduction to GNU Octave* (3rd ed., 2020), Chapter 2, Exercise 3
C	Course-created bridge or lab checkpoint

A "prompt" can be one selected part of a multi-part textbook exercise.

CLO coverage

Official course learning outcomes are assessed more than once; nothing important is a single-day checklist item.

Required outcome	Where it is developed and assessed
Interpret vectors, matrices, and complex numbers algebraically and geometrically	Weeks 1–3, 7, and 13
Solve systems by row reduction by hand and with software	Week 8; Midterm 2; cumulative final
Find 2×2 eigenpairs by hand and general eigenpairs with software	November 18 and Lab 5
Apply eigenvalues and eigenvectors	Dynamics in Week 9, explained in Week 13; Markov/normal-mode applications
Recognize triangular, symmetric, and positive-definite matrices	October 5, November 6, and December 2
Use linear combination, subspace, span, independence, basis, and dimension	Weeks 1 and 5, revisited in Weeks 8 and 11

Required outcome	Where it is developed and assessed
Write or adapt programs involving matrices, vectors, and plotting	Labs 0–7, with steadily reduced scaffolding
Apply linear algebra to real-world problems	Every lab and most class examples
Learn independently and collaboratively	Practice banks, low-stakes quizzes, paired work, lab studio, and capstone presentation
ILO/GEAR quantitative reasoning in context	Interpretation questions throughout; least-squares and capstone conclusions

Quiz dates

Friday quizzes cover **only the prior instructional week** (skip midterm week and fall break when choosing that week). Same-day lecture/lab topics are not on that day's quiz.

Quiz	Date	Focus (prior week)
Quiz 1	Fri Sep 4	Vectors, linear combinations, Lab 0
Quiz 2	Fri Sep 11	Dot products and linear functions
Quiz 3	Fri Sep 18	Norm, distance, RMS, and variation (Lecture 07)
Quiz 4	Fri Sep 25	Angle, nearest neighbors, and clustering (Lectures 08–10)
Quiz 5	Fri Oct 9	Basis, dimension, and orthonormal vectors (Lecture 14)
Quiz 6	Fri Oct 16	Matrices, Ax, and transformations (Week 7 / Lectures 18–20, Lab 3)
Quiz 7	Fri Oct 23	Systems, elimination, and RREF (Week 8 / Lectures 21–23)
Quiz 8	Fri Nov 6	Matrix multiplication, powers, and cost (Week 10 / Lectures 27–29)
Quiz 9	Fri Nov 20	Inverses, rank, and determinants (Week 11 / Lectures 30–32)
Quiz 10	Fri Dec 4	Eigenvalues, complex numbers, and Lab 5 (Week 13 / Lectures 35–37)

Not on a dedicated quiz (assessed on Midterm 2, Lab 4, Lab 6, capstone, and/or the final): Week 9 dynamics; Week 15 least squares / SVD. Use the Midterm 2 practice packet, final practice packet, and capstone practice packet as appropriate.

Exams

In-class, for the entire class period.

Exam	Date	Learning objectives
Midterm 1 (in class; not posted)	Fri Oct 2	Compute with vectors and linear functions; interpret operations; analyze distance, similarity, span, and independence; select and communicate a method. Day page: Lecture 16. Prep: study guide, practice.
Midterm 2	Fri Nov 13	Analyze matrices as transformations and models; solve and interpret systems; use products, powers, rank, inverses, and determinants; interpret computational results. Prep: study guide, practice.
Final exam	Wed Dec 16, 10:20 a.m.–12:10 p.m., FR 201	Hand fluency; select tools in new contexts; interpret software; explain connections among major concepts; apply linear algebra to quantitative problems. Prep: study guide, practice.

Project

Project	Dates	Learning objectives
Capstone	Dec 4 launch, Dec 7 studio, Dec 9 gallery	Low-rank SVD strategy; compare reconstruction error, storage, and runtime; present quantitative evidence and limitations.

Class participation

Attendance and participation in class.